

# A systematic catalogue of real three-dimensional close encounters between numbered asteroids during the Gaia DR3 window, with a measured completeness budget

D. Frascarelli<sup>1</sup>

Affiliation — TODO e-mail: dsanfra@gmail.com

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## ABSTRACT

**Context.** Close approaches between asteroids are the classical route to asteroid mass determination and a dynamical observable of the main belt. Existing encounter-based work selects individual events by hand, and no systematic catalogue of real three-dimensional asteroid–asteroid encounters has had its own completeness characterised.

**Aims.** We build a systematic catalogue of real 3D close encounters between numbered asteroids during the Gaia DR3 window (2014 July 25 – 2017 May 28) and, unlike a bulk propagation exercise, we *measure* what the detection pipeline misses rather than asserting completeness.

**Methods.** Orbits are propagated from a single frozen MPCORB snapshot centred on the window. Encounters are found by a per-timestep 3D KD-tree spatial query and refined on a dense temporal sub-grid. Reported distances use an analytic two-body refiner; a hybrid variant carries full N-body (rebound/ASSIST) refinement for the dynamically fragile (low-perihelion or high-eccentricity) subset. Perturber masses come from a joint mass+orbit least-squares fit to Gaia FPR astrometry.

**Results.** The catalogue records the minimum physical 3D separation — a genuine spatial approach, not a sky-plane co-location — for 72 236 904 candidate pairs within 0.05 AU, with encounter epoch, relative velocity, geometry, and estimated physical properties. Its distinguishing feature is a *measured* completeness budget: the Kepler-to-N-body refinement error (median  $12 \mu\text{AU}$ , p99  $2.5 \text{ mAU}$ ) and the threshold-induced false-negative rate (0.70 %, symmetric and scatter-dominated) bound this catalogue directly, while the recall of an orbital prefilter on the adverse tail (76 %) is quantified as a caveat for prefiltered runs. Applied to mass determination, the engine recovers all four calibrator masses (Ceres, Vesta, Pallas, Hygiea) at  $|z| < 3$  and determines sixteen perturbers, of which ten identifiable masses agree with recent independent work.

**Conclusions.** The catalogue is a reusable, defensibly characterised target-selection resource for asteroid dynamics and mass determination. It is explicitly not complete, and no new large–large encounter or unpublished large-body mass emerged from the numbered population.

**Key words.** minor planets, asteroids: general – celestial mechanics – astrometry – catalogs – surveys – methods: data analysis

## 1. Introduction

Close encounters between asteroids are of interest on several fronts. A slow, close approach between a massive perturber and a small test body deflects the latter’s orbit measurably, and such events are the classical route to asteroid mass determination (Michalak 2000; Goffin 2014; Fuentes-Muñoz et al. 2024, 2025). Encounters between members of the same dynamical family probe the collisional history of the belt, and the statistical distribution of mutual approach distances and velocities is itself a dynamical observable.

Existing work on encounter-based mass determination (Goffin 2014; Fuentes-Muñoz et al. 2024, 2025) starts from *hand-selected* individual encounters chosen for their favourable geometry, typically a single large perturber against a short list of well-observed test bodies. There is, to our knowledge, no systematic catalogue of real 3D asteroid–asteroid encounters whose own completeness has been characterised. That is the gap this work fills: we detect encounters exhaustively over the full numbered population in the Gaia DR3 window, and — the point that separates this catalogue from a bulk propagation exercise — we *measure* what the detection pipeline misses rather than asserting completeness.

A methodological distinction is central. Two asteroids can appear arbitrarily close in right ascension and declination as seen from Earth while remaining astronomical units apart along the line of sight. This catalogue detects the former case only when it is also the latter: it records the *minimum physical separation in three dimensions* between the two bodies’ heliocentric positions, independent of any observer. It is therefore a catalogue of genuine spatial approaches, not of sky-plane co-localisations. Light-time correction — essential when comparing model positions to Gaia’s astrometry — is deliberately *not* applied here, because the encounter is a purely geometric relation between two propagated trajectories at a common dynamical time.

The remainder of this paper describes the data and detection method (Sect. 2), the measured completeness budget that is the catalogue’s distinguishing feature (Sect. 3), the notable individual events surfaced by mining the catalogue (Sect. 4), the application to mass determination (Sect. 5), data availability (Sect. 6), and conclusions (Sect. 7).

## 2. Data and method

### 2.1. Sources and conventions

Orbital elements are taken from a single frozen MPCORB snapshot, MPCORB\_20160217.DAT (SHA-256 prefix 3e44e7d3...), selected because its osculating epoch lies at the centre of the Gaia DR3 window, minimising the propagation distance — and hence the accumulated two-body error — for Kepler refinement. The snapshot hash and download date are recorded in the catalogue’s provenance sidebar so that results are traceable to an exact input; MPCORB is a living file and results are not bit-reproducible across snapshot versions. The run is restricted to *numbered asteroids* (higher orbital quality); provisional designations are out of scope for this release.

The temporal domain brackets the Gaia DR3 Solar System survey period (Tanga et al. 2023): the propagation window is 2014 July 25 – 2017 May 28, whose start (the DR3 astrometric-resolution baseline) slightly precedes the 2014 August 5 start of nominal SSO operations, so the window fully contains the survey coverage. All internal computation is carried out in Julian Date on the Barycentric Dynamical Time (TDB) scale; conversions to/from the Barycentric Coordinate Time (TCB) that Gaia reports, and to UTC, are performed only at input/output interfaces. Distances are in AU, angles in radians, and velocities in AU/day internally, converted to user-facing units (km, degrees,  $\text{km s}^{-1}$ ) only for output.

### 2.2. Detection

The naive cost is prohibitive:  $\sim 150\,000$  bodies over a multi-year window at hour-level cadence implies of order  $10^{10}$  pairwise distances per step. Detection is therefore layered.

An *orbital prefilter* is available to prune pairs that cannot physically approach — the shipped heuristic keeps a pair only if  $|\Delta a| \leq 0.5$  AU and  $|\Delta i| \leq 30^\circ$ . This filter is a deterministic mask on osculating elements. We note here, and quantify in Sect. 3.3, that at main-belt scale ( $\gg 5000$  bodies) the pair-list prefilter is not built; the frozen run therefore applied only the KD-tree spatial query, and the  $|\Delta a|$  cut did *not* shape this catalogue (provenance `prefilter.effective = "skipped_large_n"`).

The core of the detector is a *per-timestep 3D KD-tree*. At each coarse step the heliocentric ecliptic positions of all bodies are inserted into a KD-tree ( $O(N \log N)$  construction), and a radius query returns neighbour candidates ( $O(\log N)$  per query), avoiding any  $O(N^2)$  distance matrix. The coarse cadence is  $\Delta t = 12$  h. To prevent the cadence from missing an encounter whose true minimum falls *between* two samples, the query radius is *widened* to  $r_q = 0.05$  AU +  $v_{\max} (\Delta t/2) \approx 0.0536$  AU with  $v_{\max} = 25 \text{ km s}^{-1}$ : under near-linear relative motion a pair whose separation dips below 0.05 AU between samples is within  $r_q$  at the nearer bracketing sample and therefore enters the candidate set; the sub-grid refinement then discards any whose true minimum exceeds 0.05 AU. The bracketing is exact for relative velocities up to  $v_{\max}$ , and the residual (encounters with  $v_{\text{rel}} > v_{\max}$ ) is bounded in Sect. 3.3.

Each coarse candidate is then *refined on a dense temporal sub-grid*: a  $\pm 2$  h window around the apparent minimum is sampled at 120 s, and the minimum separation, encounter epoch, and relative velocity are extracted from the refined track. Time blocks are independent, so the scan parallelises across processes.

*Propagation* is two-tiered. The coarse KD-tree scan is driven by an N-body rebound integration (WHFast; Sun + Jupiter + Saturn;  $dt = 1$  h). The sub-grid refinement that

produces the *reported* minimum distance runs through an analytic two-body Kepler propagator. Over the short Gaia DR3 arc ( $\sim 3$  yr from the central epoch) two-body propagation is adequate for the bulk of the belt, and its residual error is measured, not assumed (Sect. 3.1). For the dynamically fragile subset — pairs with perihelion  $q_{\min} < 1.8$  AU or eccentricity  $e_{\max} > 0.3$ , 8 728 509 pairs (12.08 % of the catalogue) — a *hybrid* catalogue replaces the Kepler distances with full N-body refinement (rebound IAS15; Sun + 8 planets + 4 major asteroids;  $\pm 12$  h around the Kepler minimum; 60 s sampling). The hybrid catalogue carries refinement\_method  $\in \{\text{kepler}, \text{nbody}\}$  per row together with the Kepler and N-body values and their residuals, so users needing sub-mAU geometry on high- $e$  / NEA-crossing pairs can cite it directly. Universal N-body refinement of all 72 M pairs ( $\sim 400$  CPU-days) is explicitly out of scope: the median error outside the fragile subset is already below 1 mAU (Sect. 3.1), so the marginal gain does not justify the cost.

### 2.3. Characterisation

For each detected pair the catalogue records the minimum separation and its exact epoch (from the refined sub-grid), the relative velocity at closest approach, and the geometry of the event relative to Gaia — solar elongation and an observability flag — computed on the characterised catalogue. Of the 72 236 904 encounters, 13 640 870 (18.9 %) fall in a Gaia-observable configuration. Physical properties (absolute magnitude  $H$ , and diameter and taxonomic class estimated from  $H$  with a class-dependent albedo when no direct measurement exists) are attached to both bodies. Diameters derived this way are order-of-magnitude estimates, not measurements, and any size-based selection inherits that uncertainty.

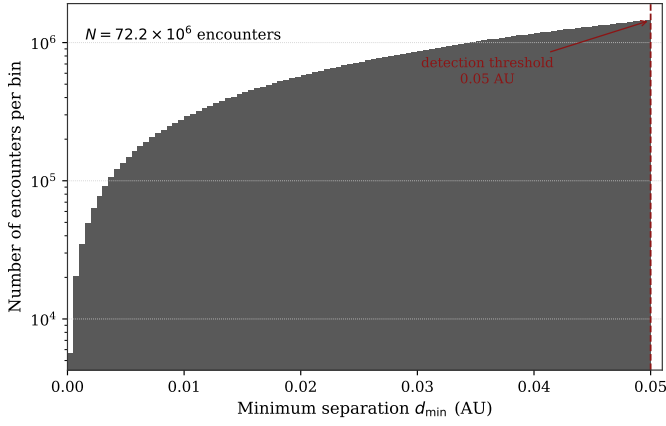
The two encountering bodies are treated as independent test particles moving in the planetary + massive-asteroid field; their *mutual* gravitational deflection over the approach is not modelled. At asteroid masses and the  $\gtrsim 10^5$  km separations that dominate the catalogue this is entirely negligible for the recorded geometry, and it is in any case the very quantity a downstream mass determination sets out to measure, not a property of the encounter itself.

### 2.4. Catalogue overview

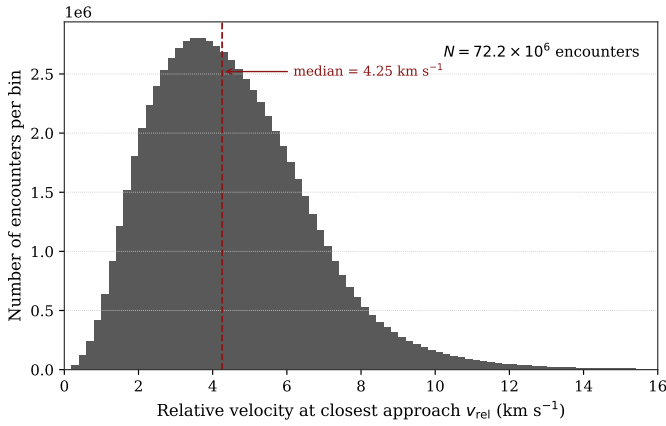
The parent population and the encounter statistics are summarised in Figs. 1–3. The minimum-separation distribution (Fig. 1) rises monotonically toward the 0.05 AU threshold, as expected for a roughly uniform spatial density of encounter geometries; the relative-velocity distribution (Fig. 2) peaks near its median of  $4.25 \text{ km s}^{-1}$ , typical of main-belt crossing velocities. The orbital distribution of the 93 010 encountering bodies (Fig. 3) reproduces the belt’s known structure — the 3:1, 5:2 and 2:1 Kirkwood gaps appear as clear voids — confirming the sample is the ordinary numbered population, not a biased subset.

## 3. Completeness budget

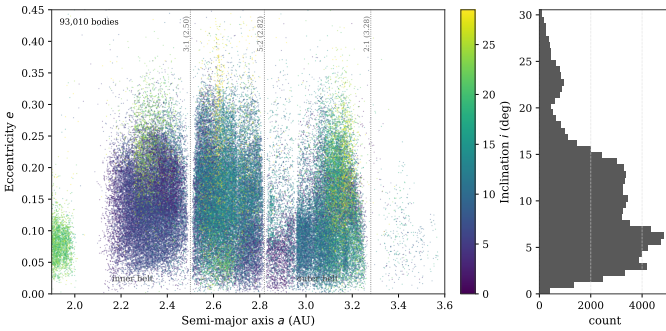
The value of this catalogue is not that it detects encounters — that is a bulk propagation — but that it states, with measured numbers, what it *misses* and how large the Kepler-vs-N-body distortion is. Three quantities are measured; each is reproducible from a dedicated script.



**Fig. 1.** Distribution of minimum 3D separations for the  $72.2 \times 10^6$  catalogued encounters (log count per bin). The dashed line marks the 0.05 AU detection threshold.



**Fig. 2.** Distribution of relative velocity at closest approach; the median is  $4.25 \text{ km s}^{-1}$ .



**Fig. 3.** Semimajor axis vs eccentricity of the 93 010 encountering bodies, coloured by inclination, with the inclination marginal at right. The 3:1, 5:2 and 2:1 mean-motion resonances (Kirkwood gaps) are marked.

### 3.1. Kepler vs N-body refinement error

The reported distances in the frozen catalogue come from the two-body refiner. We measured the resulting error against full N-body refinement in two stages. In *Stage A* (964 stratified pairs re-refined under full N-body) the median  $|\Delta d| = 12 \mu\text{AU}$ , p95 =  $678 \mu\text{AU}$ , p99 =  $2.5 \text{ mAU}$ , max =  $11.3 \text{ mAU}$ . In *Stage B* (production N-body refinement of the entire 8.73 M fragile subset) p99  $|\Delta d| = 1.99 \text{ mAU}$ , max =  $15.2 \text{ mAU}$ , with 0 failed integrations, 0 unconverged, and maximum energy drift  $5.6 \times 10^{-14}$ .

For scale,  $1 \text{ mAU} \approx 1.5 \times 10^5 \text{ km}$ , and the detection threshold is 0.05 AU. The median distortion is thus  $\sim 4$  orders of magnitude below the threshold, and even the Stage B worst case is  $\sim 30\%$  of a threshold — confined to the fragile subset, which the hybrid catalogue refines. A separate check of the truncated perturber set (the scan used only Sun + Jupiter + Saturn) shows that adding the remaining planets shifts closest-approach distances by a median  $1.3 \mu\text{AU}$  (p99  $67 \mu\text{AU}$ , max  $80 \mu\text{AU}$ ), i.e.  $\sim 100\times$  below the Kepler two-body error — the perturber truncation is not the dominant error term.

### 3.2. Threshold censoring (measured false-negative rate)

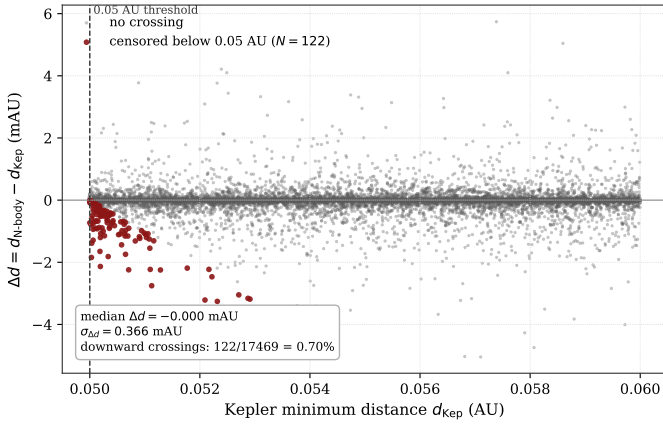
Because the catalogue contains only pairs whose *Kepler* distance is below 0.05 AU, a subtle bias arises at the threshold. When the fragile subset is re-refined to N-body, 25 283 pairs (0.29 %) cross the threshold *upward* (Kepler < 0.05, N-body  $\geq$  0.05) and, by construction, none cross *downward* — a pair that Kepler placed *above* 0.05 AU was never written, so its downward crossing *cannot be observed*. The apparent one-sided “no false negatives” is therefore censoring, not measurement.

We measured the missing side directly. Re-running Kepler detection at a widened 0.06 AU threshold over 10 000 numbered bodies — drawn to span the belt in  $(a, e, i)$  and so representative of the propagated population — and isolating the 17 469 pairs with Kepler distance in  $[0.05, 0.06) \text{ AU}$  — exactly the band a 0.05 AU catalogue discards — then re-refining each under full N-body, 0.70 % [95 % CI 0.59–0.83 %] cross *downward* below 0.05 AU (a conservative floor of  $\sim 0.42\%$  after excluding near-boundary cases). The residual  $\Delta d$  ( $N - K$ ) has median  $-3 \times 10^{-7} \text{ AU}$  and  $\sigma \approx 3.7 \times 10^{-4} \text{ AU}$  — a symmetric, scatter-dominated distribution, not a directional bias. In other words the crossing is a threshold selection effect (Eddington/Malmquist family): scatter of amplitude  $\sim 1\sigma \approx 4 \times 10^{-4} \text{ AU}$  pushes pairs just inside the threshold to just outside, and their mirror images (just outside to just inside) fall outside the catalogue. The upward rate ( $\sim 1.5\%$  in the boundary bin  $[0.045, 0.050)$ ) and the downward rate ( $\sim 0.4\text{--}0.7\%$  in  $[0.05, 0.06)$ ) are consistent with a single symmetric process. Crossings concentrate in low- $q$  / high- $e$  / fast orbits, exactly where two-body propagation is weakest — which independently validates the  $q_{\min} < 1.8 \vee e_{\max} > 0.3$  subset criterion.

Extrapolated catalog-wide (order-of-magnitude, the  $[0.05, 0.06)$  band scaling as  $N^2$ ), this implies of order  $10^5$  real < 0.05 AU encounters censored by the Kepler cut ( $\sim 1.5\text{--}2.5 \times 10^5$ ), separate from and additional to the prefilter recall deficit of Sect. 3.3. This is why the word “complete” is never used for this catalogue.

### 3.3. Prefilter recall on the adverse tail

The shipped  $|\Delta a| \leq 0.5 \text{ AU}$  prefilter is blind to eccentricity: a high- $e$  orbit reaches well inside the belt at perihelion and well outside at aphelion, so it can physically cross an orbit whose semimajor axis differs by far more than 0.5 AU. We measured the damage this would cause on the adverse subset (numbered,  $a \in [1.5, 4.0]$ ,  $e > 0.3 \vee i > 15^\circ$ ; 52 411 bodies), exploiting the fact that the prefilter is a pure deterministic mask: running detection without the prefilter and intersecting with the mask is byte-identical to running with it. Ground truth is 606 393 real adverse–adverse encounters < 0.05 AU; the prefilter keeps 463 164, for a recall of 76.38 % [95 % CI 76.27–76.49 %]. Thus 143 229 encounters would be missing from a prefiltered cata-



**Fig. 4.** Threshold censoring, measured directly. For the 17 469 pairs with Kepler distance in  $[0.05, 0.06]$  AU — the band a 0.05 AU catalogue discards — re-refined under full N-body,  $\Delta d = d_{\text{N-body}} - d_{\text{Kep}}$  is plotted against  $d_{\text{Kep}}$ . The distribution is symmetric about zero (median 0.000 mAU,  $\sigma = 0.366$  mAU); the 122 pairs (0.70 %) that cross downward below 0.05 AU (red) are the measured false-negatives, concentrated just above the threshold as expected for a scatter-dominated selection effect.

logue (a lower bound — adverse-normal pairs were not measured), and 99.6 % of that loss is the  $|\Delta a|$  cut alone. Recall degrades smoothly with eccentricity, from  $\sim 99.9$  % below  $e = 0.1$  to  $\sim 39$  % above  $e = 0.7$ .

Two points bound the impact. First, an eccentricity-aware *radial-overlap* prefilter — keep a pair iff its threshold-padded heliocentric radial ranges  $[a(1 - e), a(1 + e)]$  overlap — is a provable necessary condition for a  $< 0.05$  AU encounter, is just as cheap, and recovers 100 % recall on the adverse subset; it is the recommended replacement for any future run. Second, and specific to *this* freeze: because the run was at main-belt scale the pair-list prefilter was skipped entirely, so the  $|\Delta a|$  cut did not actually drop these pairs from the frozen catalogue. The 76 % figure measures the damage the prefilter *would* cause if applied at small  $N$ , not damage suffered here. Completeness of this freeze is instead bounded by the threshold censoring of Sect. 3.2. The 12 h coarse cadence does *not* by itself lose encounters up to  $v_{\text{rel}} = v_{\text{max}} = 25 \text{ km s}^{-1}$ , because the widened query radius (Sect. 2.2) brackets them; the only encounters it can miss have  $v_{\text{rel}} > 25 \text{ km s}^{-1}$ , and just 0.0012 % of catalogued encounters exceed that ( $p_{99.99} = 20 \text{ km s}^{-1}$ , max  $68 \text{ km s}^{-1}$ ), so the residual cadence-induced incompleteness is  $\lesssim 10^{-3}$  % — three orders of magnitude below the threshold-censoring term and negligible. For the dynamically cold bulk of the belt (low  $e$ , small  $\Delta a$ ) recall is  $\sim 99.9$  %; the incompleteness is a property of the high- $e$ /high- $i$  tail, which any science touching NEA-crossing or high- $e$  pairs must cite.

### 3.4. Recovery of literature encounter pairs

As an external check we cross-matched the hand-selected (perturber, target) pairs that Fuentes-Muñoz et al. (2025) used for mass determination against this catalogue. Of 40 176 numbered pairs drawn from their per-perturber target lists, 11 842 (29.5 %) appear as  $< 0.05$  AU encounters in the hybrid catalogue. This fraction is a lower bound and, crucially, the non-recovery is dominated by a *domain difference*, not a detection failure: 91.6 % of the unmatched pairs have both bodies present in the catalogue but never approached within 0.05 AU, because a measurable

mass deflection does not require so close an encounter and their signal-ranked target lists are therefore not restricted to 0.05 AU. A further 8.4 % are targets or perturbers outside the numbered, propagated universe (high or provisional designations), and — the decisive point — 0 % are pairs that our catalogue contains but places beyond 0.05 AU. In other words, every literature pair that is genuinely a  $< 0.05$  AU encounter and lies in the numbered population is recovered; the apparent shortfall is entirely pairs that are not  $< 0.05$  AU encounters or not numbered. Per-body recovery among the most-listed perturbers ranges from 6 % (Davida) to 61 % (Liguria), with the Ceres and Vesta calibrators at  $\sim 36$  %. Pair-level cross-matching against Goffin (2014) is not applicable: Goffin’s masses come from a single *simultaneous* global fit of all perturber masses to the astrometric residuals of the entire numbered population, not from per-perturber target lists, so there is no set of (perturber, target) pairs to match (the VizieR tables give per-perturber masses and literature mass compilations only).

## 4. Notable events

Mining the characterised catalogue surfaces individual events across four axes. We stress at the outset that the value of these tables is statistical and archival rather than a single headline discovery: there is no new large-large ( $D \gtrsim 100$  km) encounter — the two such events in the catalogue are both already known — and every candidate below is under the frozen two-body assumptions and requires N-body revalidation before publication. Distances are physical 3D separations at closest approach; diameters are  $H$ -derived estimates.

### 4.1. Large-large encounters

Only two encounters have both bodies with estimated  $D \gtrsim 100$  km, and both involve already-studied bodies (Table 1). The population of both- $D \gtrsim 50$  km encounters is richer (Table 2). The closest both- $D \gtrsim 50$  km event, (305) Gordonia  $\times$  (830) Petropolitana at  $9.9 \times 10^5$  km, involves neither of the sixteen studied perturbers and is a natural target for N-body revalidation.

### 4.2. Extreme events

The absolute closest approaches in the catalogue are between small bodies, at separations of  $\sim 1000$ – $7000$  km (Table 3). At these separations two-body propagation is most fragile, so these are precisely the events most in need of N-body revalidation; those that survive are the geometrically most striking events in the dataset.

The slowest encounters (relative velocity down to  $\sim 6$  m/s, with at least one body  $D \gtrsim 5$  km; Table 4) are the natural targets for future mass determination, since a low  $v_{\text{rel}}$  prolongs the gravitational interaction and enlarges the deflection. The combined “large + slow + close” set (one body  $D \gtrsim 50$  km,  $v_{\text{rel}} \leq 1$  km/s, dist  $\leq 10^6$  km; Table 5) is of the highest physical interest, being the configuration that maximises measurable deflection.

### 4.3. Same-region pairs (family proxy)

Pairs whose osculating elements are mutually close ( $\Delta a/a \leq 1$  %,  $\Delta e \leq 0.02$ ,  $\Delta i \leq 1^\circ$ ) and that had a physical close approach are candidate co-orbital / same-family associations. This is a proximity proxy in *osculating* elements, not a classification in proper elements, and flags rather than confirms family mem-

**Table 1.** The two encounters with both bodies  $D \gtrsim 100$  km.

| body 1    | body 2         | date (UTC) | dist (km)         | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|-----------|----------------|------------|-------------------|----------------------------|---------------|---------------|
| (1) Ceres | (57) Mnemosyne | 2017-01-10 | $6.5 \times 10^6$ | 7.25                       | 763           | 139           |
| (7) Iris  | (44) Nysa      | 2014-08-13 | $7.3 \times 10^6$ | 5.35                       | 281           | 139           |

**Table 2.** Encounters with both bodies  $D \gtrsim 50$  km, top ten by minimum distance.

| body 1           | body 2              | date (UTC) | dist (km)          | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|------------------|---------------------|------------|--------------------|----------------------------|---------------|---------------|
| (305) Gordonia   | (830) Petropolitana | 2014-09-18 | $9.88 \times 10^5$ | 3.28                       | 62.6          | 53.8          |
| (426) Hippo      | (788) Hohensteina   | 2015-04-17 | $1.15 \times 10^6$ | 8.82                       | 73.5          | 64.6          |
| (145) Adeona     | (780) Armenia       | 2016-11-09 | $1.50 \times 10^6$ | 6.34                       | 84.0          | 56.3          |
| (739) Mandeville | (415) Palatia       | 2017-04-05 | $1.70 \times 10^6$ | 8.36                       | 70.9          | 51.1          |
| (675) Ludmilla   | (80) Sappho         | 2016-02-15 | $1.71 \times 10^6$ | 2.97                       | 93.0          | 90.0          |
| (51) Nemausa     | (91) Aegina         | 2016-07-13 | $2.08 \times 10^6$ | 4.42                       | 120           | 60.6          |
| (46) Hestia      | (482) Petrina       | 2016-07-12 | $2.46 \times 10^6$ | 4.23                       | 75.6          | 60.6          |
| (758) Mancunia   | (740) Cantabria     | 2016-08-13 | $2.63 \times 10^6$ | 3.70                       | 82.9          | 53.8          |
| (739) Mandeville | (754) Malabar       | 2016-03-14 | $2.72 \times 10^6$ | 5.26                       | 70.9          | 51.6          |
| (200) Dynamene   | (1304) Arosa        | 2014-12-25 | $2.85 \times 10^6$ | 7.88                       | 79.2          | 51.3          |

**Table 3.** The five closest approaches in the catalogue.

| body 1              | body 2              | date (UTC) | dist (km) | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|---------------------|---------------------|------------|-----------|----------------------------|---------------|---------------|
| (153222) 2000 YD43  | (238587) 2004 YX3   | 2016-03-11 | 1094      | 6.90                       | 2.82          | 1.78          |
| (15072) Landolt     | (387599) 2001 XF180 | 2014-12-08 | 1760      | 5.87                       | 3.24          | 1.62          |
| (270730) 2002 QE130 | (366918) 2005 UC211 | 2016-05-29 | 2291      | 3.10                       | 1.62          | 1.18          |
| (117065) 2004 KD9   | (439086) 2011 QP5   | 2015-07-27 | 2535      | 8.25                       | 3.72          | 1.86          |
| (52249) 1981 EK21   | (408138) 2013 CL75  | 2016-11-17 | 2590      | 3.16                       | 3.39          | 1.41          |

bership. The closest such pair, (200764) 2001 XP3  $\times$  (131597) 2001 XT2, passed within 49 511 km on 2014-08-13 with nearly identical  $(a, e, i)$  —  $a \approx 2.29$  AU,  $e \approx 0.19$ ,  $i \approx 25^\circ$  for both — a plausible genetic pair worth follow-up in proper-element space.

#### 4.4. Candidate perturbers beyond the studied sixteen

Ranking large bodies ( $D \gtrsim 100$  km) *not* among the sixteen studied perturbers by their number of *useful* encounters ( $v_{\text{rel}} \leq 3$  km/s and  $\text{dist} \leq 3 \times 10^6$  km) yields the best prospects for a mass determination the frozen mass runs have not yet attempted (Table 6). (9) Metis heads the list with 37 useful events and a best pair at  $v_{\text{rel}} = 0.38$  km/s; (29) Amphitrite is the largest ( $D \approx 240$  km) with a close  $3.2 \times 10^5$  km approach. These candidates directly informed the F4 mass determination of Sect. 5.

#### 4.5. Separating new from known

A reference list of encounters touching one of the sixteen studied perturbers (1, 2, 3, 4, 7, 10, 15, 16, 31, 52, 65, 87, 88, 107, 511, 704) is provided in the companion material so that already-worked events can be separated from potential discoveries. The honest verdict is that no new large-large encounter of publication interest emerged: the belt's largest bodies are well studied, and Fuentes-Muñoz et al. (2025) already published over 230 masses (232 at  $\text{SNR} > 3$ ) covering essentially all classical large MBAs. The novelty of this work is the dataset itself — a catalogue with a measured completeness budget — and the methodological framework of Sect. 5, not a single spectacular event.

### 5. Application to mass determination

The catalogue is the target-selection front-end of an asteroid mass-determination engine. Perturber mass is obtained by a joint least-squares fit of the perturber's mass and the orbits of its test bodies — those with a  $< 0.05$  AU encounter drawn directly from the catalogue — over the full arc of Gaia FPR astrometry. The force model is ASSIST (JPL DE440 ephemeris; EIH relativistic corrections; sixteen massive asteroid perturbers); the implementation is our own, with no third-party orbit-determination dependency.

**Calibrators.** The four calibrator masses are recovered at  $|z| < 3$  against the literature (Table 7). With  $N \geq 20$  test bodies, the fit/literature ratios of Ceres (Park et al. 2016), Vesta (DAWN; Russell et al. 2012) and Hygiea (Vernazza et al. 2020) fall in  $[0.943, 0.990]$  (mean bias  $-4\%$ ); Pallas, limited to 6 encounters in the catalogue, gives ratio 1.240 ( $z = +2.67$ ).

**FOV-block covariance.** Each Gaia focal-plane crossing yields  $\sim 7$  CCD observations ( $\sim 5$  s apart) with correlated residuals. Treating them as independent underestimates  $\sigma(\text{mass})$  by a factor 1.66 (effective  $N \approx 0.36N$ ). The measured intra-crossing correlation is  $\text{ICC} = 0.32$ ; we whiten each crossing with a block covariance  $C = \text{diag}(\sigma_{\text{AL}}^2) + s_c^2 \mathbf{1}\mathbf{1}^T$ , the correlated floor  $s_c$  calibrated by bisection to  $\chi_{\text{red}}^2 = 1$ .

**Sixteen perturbers.** The engine determines all sixteen. Only (16) Psyche reaches calibrator-grade precision outside the calibrators:  $M = 2.43 \times 10^{19}$  kg,  $\sigma_{\text{stat}} = 3.3\%$ ,  $N = 36$ ,  $\chi_{\text{red}}^2 = 0.99$ , ratio 1.020 vs DE441 and 1.014 ( $z = +0.25$ ) vs the indepen-

**Table 4.** The five slowest encounters (at least one body  $D \gtrsim 5$  km).

| body 1           | body 2              | date (UTC) | dist (km)          | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|------------------|---------------------|------------|--------------------|----------------------------|---------------|---------------|
| (3791) Marci     | (110964) 2001 UW170 | 2017-01-18 | $5.85 \times 10^6$ | 0.006                      | 13.5          | 2.46          |
| (47042) 1998 WP3 | (99861) Tscharnuter | 2014-07-27 | $6.03 \times 10^6$ | 0.015                      | 5.63          | 3.24          |
| (7203) Sigeki    | (73664) 1981 EE34   | 2015-02-28 | $7.07 \times 10^6$ | 0.031                      | 8.14          | 1.48          |
| (3384) Daliya    | (309262) 2007 RZ93  | 2016-12-20 | $4.44 \times 10^6$ | 0.031                      | 6.17          | 1.23          |
| (7479) 1994 EC1  | (14954) 1996 DL     | 2017-05-12 | $7.10 \times 10^6$ | 0.032                      | 6.46          | 3.09          |

**Table 5.** The “large + slow + close” set (one body  $D \gtrsim 50$  km,  $v_{\text{rel}} \leq 1$  km/s, dist  $\leq 10^6$  km).

| body 1            | body 2              | date (UTC) | dist (km)          | $v_{\text{rel}}$<br>(km/s) | $D_1$<br>(km) | $D_2$<br>(km) |
|-------------------|---------------------|------------|--------------------|----------------------------|---------------|---------------|
| (135) Hertha      | (281202) 2007 FE47  | 2016-09-18 | $4.56 \times 10^5$ | 0.579                      | 80.3          | 1.02          |
| (371) Bohemia     | (14711) 2000 CG36   | 2016-05-14 | $5.92 \times 10^5$ | 0.982                      | 64.0          | 5.90          |
| (83) Beatrix      | (170676) 2003 YE180 | 2014-09-18 | $6.10 \times 10^5$ | 0.684                      | 65.8          | 2.14          |
| (678) Fredegundis | (200373) 2000 QH89  | 2015-03-13 | $6.27 \times 10^5$ | 0.964                      | 55.8          | 2.04          |
| (9) Metis         | (345254) 2005 UT477 | 2016-10-30 | $7.14 \times 10^5$ | 0.979                      | 197           | 1.23          |
| (639) Latona      | (10793) Quito       | 2014-12-18 | $7.27 \times 10^5$ | 0.604                      | 81.4          | 10.7          |
| (16) Psyche       | (139485) 2001 PT14  | 2015-01-19 | $9.66 \times 10^5$ | 0.869                      | 235           | 2.57          |
| (7) Iris          | (153539) 2001 SD101 | 2015-08-17 | $9.97 \times 10^5$ | 0.806                      | 281           | 2.35          |

**Table 6.** Large bodies outside the studied sixteen, ranked by useful encounters.

| body            | $D$ (km) | class | useful | best dist (km)     | min $v_{\text{rel}}$ (km/s) |
|-----------------|----------|-------|--------|--------------------|-----------------------------|
| (9) Metis       | 197      | MBA   | 37     | $7.14 \times 10^5$ | 0.378                       |
| (30) Urania     | 109      | –     | 36     | $4.40 \times 10^5$ | 0.817                       |
| (40) Harmonia   | 141      | –     | 36     | $6.58 \times 10^5$ | 1.028                       |
| (19) Fortuna    | 133      | MBA   | 32     | $7.43 \times 10^5$ | 0.678                       |
| (21) Lutetia    | 120      | –     | 30     | $8.18 \times 10^5$ | 0.609                       |
| (64) Angelina   | 104      | MBA   | 30     | $8.73 \times 10^5$ | 0.992                       |
| (44) Nysa       | 140      | –     | 30     | $8.95 \times 10^5$ | 0.961                       |
| (20) Massalia   | 178      | –     | 28     | $8.78 \times 10^5$ | 0.514                       |
| (29) Amphitrite | 240      | –     | 27     | $3.17 \times 10^5$ | 0.643                       |
| (27) Euterpe    | 141      | –     | 26     | $4.63 \times 10^5$ | 0.847                       |

**Table 7.** Calibrator masses recovered by the joint orbit+mass fit. The  $z$  column uses the total (formal  $\oplus$  systematic)  $\sigma$ .

| body   | $N$ | fitted mass (kg)      | $\sigma_{\text{tot}}$ | ratio fit/lit | $z$   |
|--------|-----|-----------------------|-----------------------|---------------|-------|
| Ceres  | 28  | $8.96 \times 10^{20}$ | 4.6 %                 | 0.955         | −1.01 |
| Vesta  | 28  | $2.44 \times 10^{20}$ | 4.6 %                 | 0.943         | −1.30 |
| Hygiea | 20  | $8.22 \times 10^{19}$ | 5.7 %                 | 0.990         | −0.13 |
| Pallas | 6   | $2.54 \times 10^{20}$ | 7.0 %                 | 1.240         | +2.67 |

dent Fuentes-Muñoz et al. (2025) FPR fit. Six perturbers whose deflection sits below the per-encounter astrometric noise return ratios in  $[0.39, 0.72]$ : this is the mass↔orbit degeneracy (the fit reproduces the astrometry with a smaller mass and a re-adjusted orbit — regression toward zero mass), and for these the formal Fisher  $\sigma$  underestimates the error.

External  $\sigma$  by jackknife. Because the formal  $\sigma$  scales as  $1/\sqrt{N}$  and ignores the mass↔orbit regression seen only by leaving encounters out, we report a jackknife  $\sigma$  that reincorporates it. It is systematically 2–3× the formal  $\sigma$ . Its diagnostic value is visible in the calibrators themselves: Pallas and Vesta — whose true masses are known — sit at  $|z| > 3$  under the formal  $\sigma$  but within  $|z| < 3$  under the jackknife  $\sigma$ , showing the formal bar is genuinely too tight rather than the jackknife being permissive. Cross-checking the ten *measured* (identifiable,  $\text{snr}_{\text{jack}} \geq 3$ ) perturbers against Fuentes-Muñoz et al. (2025),

10/10 fall within  $|z| < 3$  under the jackknife  $\sigma$ , versus 5/10 under the formal  $\sigma$  (6/6 vs 3/6 for the non-calibrators, the genuinely independent comparison). The remaining perturbers are flagged `not_identifiable` — their deflection does not clear the per-encounter noise and their masses are reported as explicit bounds, not measurements.

Beyond the sixteen. The engine is not restricted to the sixteen ephemeris perturbers: each perturber is integrated as a massive rebound particle from its own orbital elements, so a seventeenth body outside sb441–n16 is added as a seventeenth massive particle with no double-counting (the ASSIST ASTEROIDS force is deliberately excluded). This was demonstrated on (19) Fortuna — orbit fixed from JPL Horizons, mass free — which fits with  $\chi^2_{\text{red}} = 0.977$  and  $M = 1.13 \times 10^{19} \pm 2.2 \times 10^{18}$  kg. The methodological gate is green: the engine generalises to bodies beyond the ephemeris set. Fortuna is not itself a new mass (it appears in Goffin 2014 and Fuentes-Muñoz et al. 2025), and its 20 %  $\sigma$  makes it a coarse determination consistent with the literature within the jackknife bar ( $z = +1.25$ ). The catalogue’s top ranked out-of-sixteen candidate, (9) Metis (37 useful encounters, Sect. 4.4), was run as a second out-of-sixteen case and fits with  $\chi^2_{\text{red}} = 0.981$  and  $M = 4.74 \times 10^{18} \pm 1.53 \times 10^{18}$  kg (55 targets, jackknife  $\sigma = 32$  %), a ratio of 0.73 against Fuentes-Muñoz et al. (2025) ( $6.48 \times 10^{18}$  kg), consistent within the jackknife bar ( $z = -1.14$ ). Fortuna and Metis thus scatter to either side of the literature (+30 % and −27 % in central value)



but both remain consistent once the honest external  $\sigma$  is used — confirming the method generalises while underscoring that a single out-of-sixteen perturber with  $\sim 50$  Gaia-FPR targets yields only a coarse (20–30 %) determination, not a competitive mass. Neither is a new mass: both are already in Fuentes-Muñoz et al. (2025).

We tested whether the mean  $-4\%$  calibrator bias is an artefact of an incomplete perturber background by enlarging it from the sixteen ephemeris bodies to thirty-five — adding the twenty most massive asteroids of Fuentes-Muñoz et al. (2025) ( $\approx 1.6 \times 10^{20}$  kg spread across the belt), with masses from their Table 5 and orbits from JPL Horizons, using the same custom-perturber machinery. The calibrator masses shift by  $< 0.25\%$  (Ceres  $-0.02\%$ , Vesta  $-0.23\%$ , Hygiea  $+0.18\%$ ), three orders of magnitude below the deficit being probed; the systematic floor  $f_{\text{sys}}$  does not drop ( $4.16\% \rightarrow 4.26\%$ ). Background incompleteness is therefore *not* the origin of the  $-4\%$  bias — a far-field, dispersed perturber population averages to nearly zero deflection over the target ensemble. The residual bias is more plausibly attributable to per-target orbit imperfection over the short Gaia arc or unmodelled per-encounter astrometric systematics.

## 6. Data availability

The primary product is the Kepler-refined candidate catalogue, `encounters_catalog_rebound_005au.parquet` (72 236 904 rows, SHA-256 `b0272be7...`), a typed Parquet file with a machine-readable provenance sidecar recording the MPCORB snapshot hash, the frozen configuration, and the pipeline parameters. The derived *hybrid* catalogue, `encounters_catalog_hybrid_stageb.parquet`, carries per-row refinement `method`  $\in \{\text{kepler}, \text{nbody}\}$  and both the Kepler and N-body distances, epochs, velocities, and their residuals; it should be cited for any science depending on sub-mAU geometry. The characterised catalogue, `encounters_characterized_full.parquet`, adds observability, solar elongation, magnitudes, diameters, and taxonomic classes for the full 72 M rows. Anything derived from the catalogue must cite both the freeze (catalog SHA `b0272be7...`) and the code commit at its own generation time.

## 7. Conclusions

We have built and characterised a systematic catalogue of 72 236 904 real 3D close encounters ( $< 0.05$  AU) between numbered asteroids during the Gaia DR3 observation window. Unlike prior encounter-based work, which selects individual events by hand, the catalogue is exhaustive over the numbered population and — its distinguishing feature — comes with a *measured* completeness budget rather than an assumed one: the two-body refinement error (median  $12 \mu\text{AU}$ , p99  $2.5 \text{ mAU}$ ), the threshold-induced false-negative rate ( $0.70\%$ , symmetric and scatter-dominated, implying  $\sim 10^5$  censored encounters catalogue-wide), and the orbital prefilter recall on the adverse high- $e$ /high- $i$  tail ( $76\%$ , with a zero-cost radial-overlap fix that recovers  $100\%$ ). A hybrid variant provides N-body-grade geometry on the dynamically fragile subset.

The catalogue is a working target-selection front-end for mass determination: a joint orbit+mass engine on Gaia FPR astrometry recovers all four calibrators at  $|z| < 3$ , determines sixteen perturbers (all ten identifiable masses consistent with Fuentes-Muñoz et al. 2025 at  $|z| < 3$  under a jackknife  $\sigma$ ), and generalises to bodies beyond the ephemeris set. We are explicit

about the limits: distances are candidates under frozen two-body assumptions except on the N-body-refined subset, the catalogue is not complete, and no new large-large encounter or unpublished large-body mass emerged — the belt's massive bodies are already well surveyed. The contribution is the dataset with its quantified budget and the methodological framework (FOV-block covariance, identifiability criterion, external jackknife  $\sigma$ ), which together turn a bulk propagation into a defensibly characterised catalogue.

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